ShareLoop

Recycling & Item Sharing Platform

Give What You Don't Need. Get What You Do.

PROJECT FEATURE DOCUMENTATION

Undergraduate IT Project Proposal • Software Requirements Overview

Theme: Sustainability • Community • Technology

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1. Project Overview

ShareLoop is a community-driven mobile and web platform that enables individuals to give away items they no longer need and request items they require, instead of purchasing new products or discarding usable belongings. The platform connects two everyday situations that normally never meet each other: a household with an unused item taking up space, and a nearby person who genuinely needs that same item but cannot afford or find it easily.

The core problem ShareLoop addresses is the disconnect between waste and need. Every day, usable clothes, books, school supplies, electronics, and household goods are thrown away or left unused simply because there is no convenient way to pass them on to someone who could benefit from them. At the same time, students, low-income families, and individuals in need often struggle to access these same essential items. ShareLoop closes this gap through a simple, safe, and structured digital exchange system.

How ShareLoop Promotes Recycling and Reuse

- Gives unused items a second life instead of ending up in landfills.
- Encourages users to think of donation as the first option before disposal.
- Extends the usable lifespan of clothing, electronics, books, and household goods.
- Reduces demand for newly manufactured products, indirectly lowering resource consumption.

Connecting Givers and Requesters

Users can act as donors by posting items they wish to give away, or as requesters by posting items they need along with a short explanation of why they need them. The platform then facilitates a transparent selection and handover process so that items reach the people who need them most.

Reducing Waste and Building a Sharing Culture

By making reuse as convenient as buying something new, ShareLoop shifts everyday behaviour toward sustainability. Over time, repeated positive exchanges, community recognition, and gamified contribution tracking build a culture where sharing becomes a normal and rewarding part of daily life rather than an occasional act of charity.

Core Idea: ShareLoop turns “I don’t need this anymore” into “someone nearby needs exactly this” — reducing waste while directly supporting the community.

2. Main Functional Features

This section describes the core features that allow users to give away, request, and locate items on the ShareLoop platform.

2.1 Give Away Posts

Users can create a post for any unused item they wish to donate, including clothes, shoes, books, school supplies, electronics, and household items. Each post is designed to give the requester enough information to decide whether the item suits their need.

- **Upload Photos** — clear images of the item’s actual condition.
- **Item Details** — name, description, and condition (new, good, used).
- **Category Selection** — assigns the item to the correct browsing category.
- **Location Tagging** — district and city so nearby users can find it easily.
- **Publish** — the post becomes visible to all users searching in that category or area.

2.2 Item Requests

Users who need something can create a request post rather than waiting for a matching donation to appear. Typical examples include school books, A/L notes, school supplies, clothes, shoes, and household items.

- **Description** — what exactly is needed (e.g. “Grade 10 science textbook”).
- **Reason** — a short explanation of why the item is needed, shown to potential donors.

This reason field is important: it allows donors to understand the requester’s situation and make a fair, informed decision when several people request the same item.

2.3 Location & District Filtering

Because item pickup usually happens in person, location is central to a good user experience. ShareLoop allows users to filter both give-away posts and requests by district, city or town, and proximity to their own location.

- Users can browse items available only within their own district.
- Nearby results reduce travel distance and make collection practical.
- Donors can prioritise requesters who are geographically closer.

This location-based filtering turns a nationwide platform into a series of practical, local exchanges, which is essential for a system built around physical items rather than digital goods.

3. Engagement & Motivation

To keep users active and encourage continuous contribution, ShareLoop includes gamification features that reward generosity and recognise contributors.

3.1 Gamification — Badges & Levels

Users earn badges as they donate more items, creating a visible sense of progress and achievement within their profile.

Badge	Items Donated	Recognition Level
Bronze	5 items donated	New Contributor
Silver	10 items donated	Active Contributor
Gold	25 items donated	Trusted Contributor
Platinum	50+ items donated	Community Champion

This badge progression encourages users to keep contributing rather than stopping after a single donation. Visible achievement status also builds trust, since other users can see a donor's history of positive activity on the platform.

4. Usability & Accessibility Features

ShareLoop is designed to be simple to navigate and accessible to users from different linguistic and technical backgrounds across Sri Lanka.

4.1 Category-Based Browsing

Items and requests are organised into clear categories so users can search and browse efficiently rather than scrolling through unrelated listings.

- Educational Books
- School Supplies
- Clothes
- Shoes
- Electronics
- Household Items
- Other

Category-based browsing significantly reduces search time and helps users quickly reach the type of item they are looking to give or receive.

4.2 Multi-Language Support

The application supports Sinhala, Tamil, and English, allowing users from different communities across the country to use ShareLoop comfortably in their preferred language.

Multilingual support removes a major barrier to adoption in a multi-ethnic society, ensuring the platform's social impact is not limited to English-speaking users alone.

5. User Roles

ShareLoop defines four primary roles that structure how users interact with the platform. A single individual is not restricted to one role — the same person can act as a donor in one exchange and a requester in another.

Normal User — Any registered individual. A normal user can browse listings, search by category or location, and switch between donor and requester roles as needed.

Donor — A user who creates give-away posts. Donors upload item details, review incoming requests along with each requester's stated reason, select the most suitable requester, and coordinate handover.

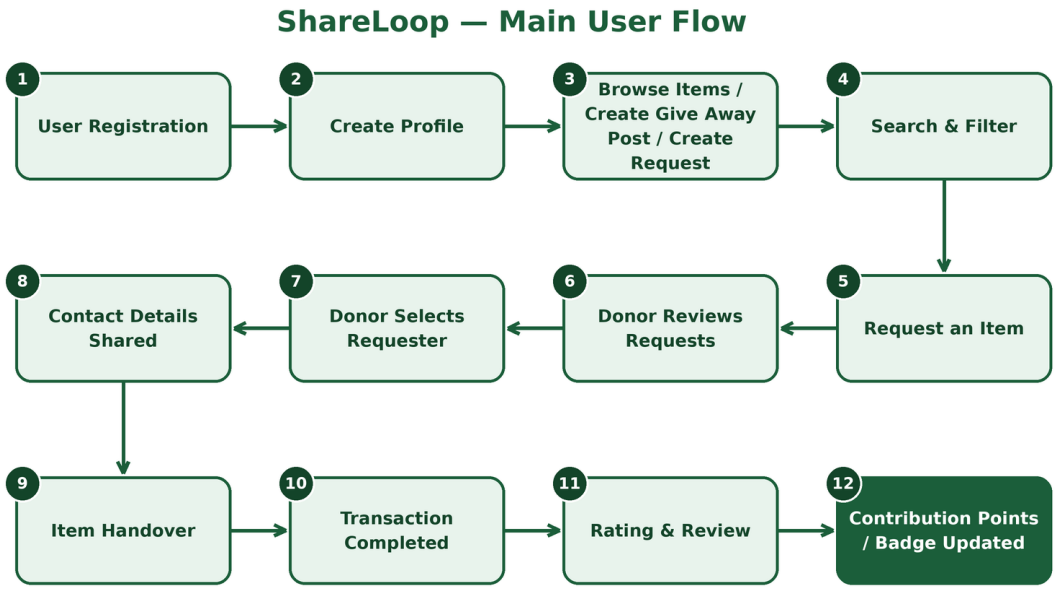
Requester — A user who creates or responds to item requests. Requesters describe the item they need, explain why they need it, and arrange collection once approved.

Administrator — Responsible for overall platform integrity: moderating users and posts, managing categories, and ensuring the community guidelines are enforced consistently.

Note: A single user account can act as both a Donor and a Requester depending on the situation — there is no separate registration for each role.

6. Main User Flow

The flow below outlines the end-to-end journey a user follows, from initial registration through to the completion of an item exchange and the resulting contribution rewards.



1. User Registration
2. Create Profile
3. Browse Items / Create Give Away Post / Create Request
4. Search & Filter
5. Request an Item
6. Donor Reviews Requests
7. Donor Selects Requester
8. Contact Details Shared
9. Item Handover
10. Transaction Completed
11. Rating & Review
12. Contribution Points / Badge Updated

7. Feature Summary Table

The table below consolidates the eight core features of ShareLoop for quick reference during review and presentation.

#	Feature	Description	Main Benefit
1	Give Away Posts	Post unused items with photos, details, category, and location.	Gives unused items a new home instead of disposal.
2	Item Requests	Post a request describing a needed item and the reason for it.	Lets users express need directly instead of only browsing.
3	Location & District Filtering	Filter items and requests by district, city, or proximity.	Makes collection practical and localised.
4	Gamification	Bronze to Platinum badges based on	Motivates continuous

#	Feature	Description	Main Benefit
	(Badges & Levels)	donation count.	contribution.
5	Donor Selection & Reason System	Donor reviews requester reasons before approving.	Ensures fair, informed item allocation.
6	User Ratings & Reviews	Star rating and review after each exchange.	Builds a trustworthy user reputation system.
7	Category-Based Browsing	Items organised into clear, searchable categories.	Speeds up and simplifies searching.
8	Multi-Language Support	Sinhala, Tamil, and English language options.	Improves accessibility across communities.

8. Benefits of the System

ShareLoop's impact extends across environmental, social, economic, and educational dimensions, reflecting its design as a holistic community sustainability platform.

🌱 Environmental Benefits

- Reduces waste sent to landfills.
- Encourages reuse over disposal.
- Supports broader recycling and circular-economy goals.
- Reduces unnecessary consumption of newly manufactured goods.

🤝 Social Benefits

- Helps people directly access essential items they need.
- Encourages a culture of community sharing and generosity.
- Supports students and low-income households in particular.
- Builds trust between users through transparent exchanges.

💰 Economic Benefits

- Reduces unnecessary purchasing of new items.
- Helps users obtain useful items free of charge.
- Eases financial pressure on students and low-income families.

📚 Educational Benefits

- Encourages responsible, mindful consumption habits.
- Raises awareness about sustainability among young users.
- Promotes a sharing culture within schools and university communities.

9. Conclusion

ShareLoop brings together technology, sustainability, and social responsibility into a single, practical platform. By making it as easy to give away an item as it is to buy one, the system directly tackles two everyday problems at once: unnecessary waste and unmet community need.

Through structured give-away and request posts, location-based filtering, and a donor-approval process that lets donors choose based on real needs, ShareLoop provides a fair and reliable environment for physical item exchange. Gamification through badges keeps users engaged and rewards consistent generosity.

Ultimately, ShareLoop aims to make recycling, reuse, and generosity a natural and rewarding part of everyday life.

Vision: A community where nothing usable goes to waste, and everyone in need has a place to turn.